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THE PRESENT AND FUTURE

JACC STATE-OF-THE-ART REVIEW

Sudden Cardiac Death in Young Athletes

JACC State-of-the-Art Review

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ABSTRACT

Athletes epitomize the healthiest segment of society. Despite this premise, sudden cardiac death may occur in apparently healthy athletes, attracting significant attention not only in the medical community but also in laypersons and media. The incidence of sudden cardiac death is variably reported, and epidemiological burden differs among cohorts. Athletes appear to be at risk of developing fatal arrhythmias when harboring a quiescent cardiac disorder. Primary cardiomyopathies, ion channelopathies, and coronary artery anomalies are prevalent causes in young individuals. Cardiac assessment of athletes can be challenging because these individuals exhibit a plethora of electrical, structural, and functional physiological changes that overlap with cardiac pathology. A diagnosis of cardiac disease in a young athlete is not necessarily an indication to terminate competition and sports participation. International guidelines, traditionally focused on disqualification of individuals with cardiac disease, have recently adopted a more liberal attitude, based on a careful assessment of the risk and on a shared-decision making approach. (J Am Coll Cardiol 2024;83:350–370)
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Sudden cardiac death (SCD) may occur in young and apparently healthy individuals, including athletes.[1-3] The benefits of exercise on reducing the burden of atherosclerotic coronary artery disease are irrefutable, and athletes often represent the healthiest segment of society.[4] Despite these premises, an abnormal structural or electrical cardiac substrate may promote fatal arrhythmias during adrenergic stimuli, especially in individuals who exercise vigorously. Inherited cardiac conditions are the predominant cause of SCD in young (<35 years of age) athletes, and SCD may be the first manifestation of concealed disease in asymptomatic individuals.[1,5] The diagnosis is often made only through a detailed postmortem examination, which is crucial in elucidating the cause of death[6,7] and governing clinical assessment of surviving first-degree relatives. The counterintuitive nature of these tragedies draws considerable attention to the lay and scientific community. SCD in athletes may be prevented through the implementation of policies aimed at the early identification of cardiac conditions that pose a risk in asymptomatic individuals (preparticipation screening) and strategies focused on increasing the likelihood of successful resuscitation of cardiac arrests.[8,9]

This review focusses on the epidemiology and the causes of SCD in young athletes, and provides insight into the main controversies and challenges in the identification and subsequent management of athletes with cardiac disease.

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EPIDEMIOLOGY

The incidence of SCD in athletes is extremely variable and reflects the demographic heterogeneity of the cohorts studied. Age, sex, ethnicity, and sporting discipline are important determinants of risk.

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HIGHLIGHTS

- The incidence of SCD in young athletes has varied widely across studies. Inherited cardiac conditions and congenital abnormalities, such as coronary artery anomalies, are the most common etiologies of SCD in young athletes
- Early identification of disease in asymptomatic individuals through preparticipation screening and policies aimed at implementing cardiopulmonary resuscitation in public spaces are means to prevent these events.
- A diagnosis of cardiac disease should not necessarily disqualify young athletes from participation in sports. Expert assessment and shared decision making should guide exercise prescription and engagement even in high-intensity sports.

Moreover, a unanimous agreement on the definition of "athlete" does not exist, and individual studies interpret the term differently with respect to the amount, type, or intensity of exercise. Furthermore, some studies have included only sudden cardiac deaths while others also report sudden cardiac arrests (SCA) (Table 1).

Robust data on incidence of SCD in sport are derived from a 10-year review of a National Collegiate Athletic Association (NCAA) database (athletes of 17-24 years of age) in the United States. In this cohort, SCD was the most common medical cause of death, with an incidence of 1.9 per 100,000 athlete-years.[10] A prospective assessment of SCDs within the Veneto region in Italy showed an incidence of 3.6 per 100,000 athlete-years before preparticipation screening policies were introduced in the early 1980s with a significant decline to 0.4 per 100,000 after these policies were fully implemented.[9] The incidence of SCD in Minnesota high school athletes (age 12-19 years) over 19 academic years was 0.24 per 100,000 athlete-years.[11] In contrast, a recent study of a large cohort of screened adolescent football players with an accurately defined numerator and denominator revealed an incidence of SCD of 6.8 per 100,000 athlete-years.[12] The incidence of SCD and SCA is higher in older (age >35 years) athletes compared with younger counterparts. According to a Danish national study,[13] the estimated incidence of SCD was 0.47 to 1.21 per 100,000 person-years in young (age ≤35 years) competitive athletes compared with 6.64 per 100,000 person-years in older (age >35 years) athletes.

Athletes of African or Afro-Caribbean origin often excel in sport competitions, especially in certain disciplines such as basketball, soccer, and marathon running. A recent prospective study, based on the National Centre for Catastrophic Sports Injury Research registry in the United States, reported on 331 confirmed cases of SCA and SCD (158 survivors, 173 fatalities); the annual incidence rate among African American male NCAA Division I basketball players was 1:2,087 athlete-years and was the highest compared with other ethnic groups.[14]

For many years, competitive sport has been dominated by men, but the last 2 decades have witnessed an increasing number of women participating in high-level competition. A pooled analysis of 3 large databases showed that cardiac arrest during sport appears to be >10-fold lower in women compared with men, despite similar subject characteristics, including age, history of cardiac disease, and location of occurrence.[15] Variables implicated in this difference may include the effect of sex on physiological cardiac adaptation to exercise and chamber remodeling, the higher prevalence of myocardial fibrosis in men, and hormonal factors.[16,17]

ETIOLOGY

SCD in young athletes is caused by a diverse spectrum of cardiac diseases; however, inherited cardiac conditions are predominant in this age group. Cardiac autopsy may elucidate structural abnormalities, but the heart may be structurally normal in this setting, indicating sudden arrhythmic death syndrome, where a primary arrhythmia syndrome is the most likely underlying etiology[18,19] (Figure 1).

THE VALUE OF AN EXPERT CARDIAC PATHOLOGY ASSESSMENT. The universal declining autopsy rate is demonstrated in a recent FIFA report on sudden death in young footballers, which showed that of 617 cases from 67 countries, only 127 had an autopsy.[20] A recent review within Europe highlighted that up to 40% of SCDs in young individuals (<50 years of age) did not have an autopsy.[21] A detailed cardiac autopsy is crucial, especially in young SCDs where the root cause is hereditary, with implications for family members[18] (Figure 2). An assessment by expert cardiac pathologists using a standardized protocol

ABBREVIATIONS AND ACRONYMS

ACM = arrhythmogenic cardiomyopathy
AED = automatic external defibrillator
CAA = coronary anomaly
CMR = cardiovascular magnetic resonance
DCM = dilated cardiomyopathy
HCM = hypertrophic cardiomyopathy
LV = left ventricle/ventricular
LVH = left ventricular hypertrophy
MB = myocardial bridge
RV = right ventricle/ventricular
SCA = sudden cardiac arrest
SCD = sudden cardiac death

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TABLE 1 Epidemiology of SCD and SCA in Athletes

First Author	Year	Country	Case Identification	SCD/SCA	Years Studied	Annual Incidence	Number of Years	Age Range, y	Number of Deaths
Van Camp et al[113]	1995	United States	National Center for Catastrophic Injury Research and media database	SCD	1983-1993	0.3:100,000	10	17-24	160
Corrado et al[25]	2003	Italy	Mandatory death reporting	SCD	1979-1999	2.3:100,000	20	12-35	55
Maron et al[2]	2009	United States	USA Registry for Sudden death in Athletes	SCA + SCD	1980-2006	0.6:100,000	27	8-39	690
Drezner et al[102]	2009	United States	1,710 high schools with AEDs surveyed for SCA or SCD	SCA + SCD	2006-2007	4:100,000 SCA + SCD 2.2:100,000 SCD	Within 6 months of survey	14-17	14
Harmon et al[114]	2011	United States	Parent Heat Watch database, NCAA Resolutions list, insurance claims	SCD	2004-2008	2.3:100,000	4	18-26	37
Marijon et al[115]	2011	France	Emergency medical response to confirmed cardiac cases	SCA + SCD	2005-2010	1:100,000	5	12-75	820
Steinvil et al[100]	2011	Israel	Retrospective review (2 Israeli newspapers by 2 media researchers)	SCD	1985-2009	1st 2.5:10,000 2nd 2.6:100,000	24	12-44	24
Roberts et al[11]	2013	United States	Minnesota State High School League records	SCD	1993-2012	0.24:100,000	19	12-19	4
Maron et al[116]	2013	United States	USA Registry for Sudden Death in Athletes and NCAA resolutions list for cardiac cases	SCD	2002-2011	1.6:100,000 presumed 1.2:100,000 confirmed	9	17-26	64
Maron et al[99]	2013	United States	USA Registry of Sudden Death in Athletes	SCD	1986-2011	0.6:100,000	26	12-18	13
Toresdahl et al[117]	2014	United States	2,149 school students—events occurred on school campus	SCA + SCD	2009-2011	1.1:100,000	2	14-18	18 SCA 2 SCD
Harmon et al[3]	2015	United States	NCAA database	SCD	2003-2013	1.9:100,000	10	17-24	79
Marijon et al[118]	2015	United States	Population the Oregon-Sudden Unexpected Death Study (Oregon-SUDS)	SCA	2002-2013	2.2:100,000 (during sports activities)	11	35-65	1,247 SCA cases, 63 SCA during sports activities
Malhotra et al[12]	2018	United Kingdom	Football association registry	SCD	1996-2016	6.8:100,000	20	16	8
Peterson et al[14]	2021	United States	National Center for Catastrophic Sports Injury Research in collaboration with national sports organizations	SCD + SCA	2014-2018	1.5:100,000 (high school) 1.9:100,000 (NCAA level)	4	11-29	331 SCA, 173 SCD

NCAA = National Collegiate Athletic Association; SCA = sudden cardiac arrest; SCD = sudden cardiac death.

including histology has been shown to improve the accuracy of postmortem cardiac diagnoses in this setting.[1,6]

STRUCTURAL ABNORMALITIES. Hypertrophic cardiomyopathy. Hypertrophic cardiomyopathy (HCM) is the most common inherited cardiac disease, with a prevalence of 0.2% in the general population. It is defined by unexplained left ventricular hypertrophy (LVH) in the absence of abnormal loading conditions.[22] The electrocardiogram (ECG) remains crucial in the diagnosis and is abnormal in >90% of cases.[23] The echocardiogram, besides the demonstration of LVH, may reveal left ventricular (LV) outflow obstruction, which is reported either at rest or during stress, in two-thirds of patients.[24] More detailed imaging tests, such as cardiovascular magnetic resonance (CMR), provide a more accurate quantification of the magnitude and pattern of LVH and the detection of concomitant myocardial fibrosis. HCM has been reported to be the cause of SCD in 2% to 36% of athletes,[1,2,25] and some studies from the United States indicate that it is the predominant cause of SCD in young athletes.[2] Prevention of SCD relies on the use of implantable cardioverter-defibrillators (ICDs) in primary and secondary prevention, after accurate prognostic stratification. From a pathology standpoint, the diagnosis of HCM is made by the detection of increased heart weight and/or asymmetric LVH on macroscopic exam and by the presence of widespread myocyte disarray at histology.[26]

Arrhythmogenic cardiomyopathy. Arrhythmogenic right ventricular cardiomyopathy (ARVC) is an inherited condition that is characterized by progressive

fibrofatty replacement of the myocardium and a predilection for fatal arrhythmias. The disease is generally caused by mutations in genes encoding cardiac desmosomal proteins, with a prevalence of 1:1,000 in the general population.[27] ARVC appears to be a major cause of SCD in athletes, as shown by studies from the Veneto region in Northern Italy.[25] This condition appears to be a predominant etiology in athletes who die during exercise. The diagnosis of ARVC is complex and traditionally based on the Task Force Criteria, which include a series of clinical, electrocardiographic, structural, and functional traits.[28] Patients at risk should be implanted with an ICD, although it is often challenging to identify individuals who would benefit from a device, especially for primary prevention. ARVC has been increasingly shown to be a biventricular disease including isolated LV involvement in some cases. These observations have resulted in calls for revising the existing diagnostic criteria as well as the nomenclature of the condition, with proposals to use the term "arrhythmogenic cardiomyopathy (ACM)."

At autopsy, the diagnosis is made by the presence of fibrosis in a subepicardial distribution that may be associated with fat in the absence of significant coronary artery disease.[29]

Dilated cardiomyopathy. Dilated cardiomyopathy (DCM) is defined by the presence of LV or biventricular dilatation and systolic dysfunction in the absence of abnormal loading conditions or coronary artery disease sufficient to cause global systolic impairment.[30] DCM has a propensity to result in serious ventricular arrhythmias that may characterize the natural history of the disease, especially in some genetically mediated forms (pathogenic variants in LMNA, PLN, FLNC, DSP), which often overlap with ACM.[31]

DCM is a relatively rare cause of SCD in athletes (1%-3% of cases)[1,2,10] because most individuals with this condition have impaired functional capacity and are unable to engage in vigorous exercise for prolonged periods. At autopsy, the heart is characterized by an increased weight with dilation and wall thinning of the left ventricle[32] and replacement fibrosis of LV myocardium.

Myocarditis. Myocarditis is an inflammatory disease of the myocardium and demonstrates heterogeneity in its etiology and clinical presentation.[33] Viral infection is the most common cause, although autoimmune mechanisms such as in giant cell and eosinophilic myocarditis are also recognized.[33] Myocarditis accounts for a range of 2% to 9% of cases of SCD in athletes,[1,2] and fatal arrhythmias may be the first manifestation of the disease. CMR is the gold-standard noninvasive test, and endomyocardial biopsy may provide a confirmation of the diagnosis and insights on the histological type. Postmortem diagnosis is often challenging and relies on the detection of a significant lymphocytic infiltrate and myocyte necrosis. The natural history of myocarditis is highly variable, with most patients recovering fully and others developing DCM with symptoms of heart failure.[34] Some cases show a propensity to ventricular arrhythmias especially in the acute phase. Athletes with myocarditis should refrain from intense exercise for at least 3 months from the acute event.[35]

Idiopathic LVH. Idiopathic LVH is a fairly novel entity that is increasingly reported following postmortem examination in cases of SCD.[36,37] Recent studies[2,3,38] reveal that in selective cohorts, idiopathic LVH may be reported in up to 30% of victims of SCD. Idiopathic LVH is characterized by unexplained LVH (heart weight >500 g in men and >400 g in women or increased heart weight as a function of body weight[39] and LV wall thickness >15 mm)[40] in the absence of the myocardial disarray. A recent study investigated whether idiopathic LVH and familial HCM are part of the same disease, performing comprehensive assessment of first-degree family members of 46 decedents with idiopathic LVH. Of the 125 family members investigated, none had features compatible with HCM, suggesting that idiopathic LVH is a distinct disease entity.[41]

Idiopathic fibrosis. Myocardial fibrosis is a common feature of many cardiac conditions. In approximately 8% of cases, myocardial fibrosis is the only pathological feature detected at cardiac autopsy among athletes who are victims of SCD.[1] A recent study showed that a 10% of subjects with idiopathic fibrosis at autopsy had actionable variants in genes associated with ACM, DCM, and HCM without autopsy findings of those diseases; hence, it is likely that myocardial fibrosis may precede the overt phenotypic expression of a cardiomyopathy, but still constitute a substrate of fatal arrhythmias.[42]

Coronary artery anomalies. Coronary artery anomalies (CAAs) have been reported as the second most frequent cause of SCD in young athletes in the United States accounting for 17% of deaths.[2] A study from the United Kingdom reported CAAs in 5% of cases in athletes who died suddenly,[2] and another recent study attributed 9% of all SCDs in adolescent athletes to CAAs.[43] Anomalous origin of the coronary artery from the wrong sinus with an inter-arterial course between the pulmonary artery and aorta are implicated in SCD.[44] The inter-arterial course may also have an intramural segment running within the aortic wall.[43] Imaging tests such as CMR and computed tomography coronary angiography provide

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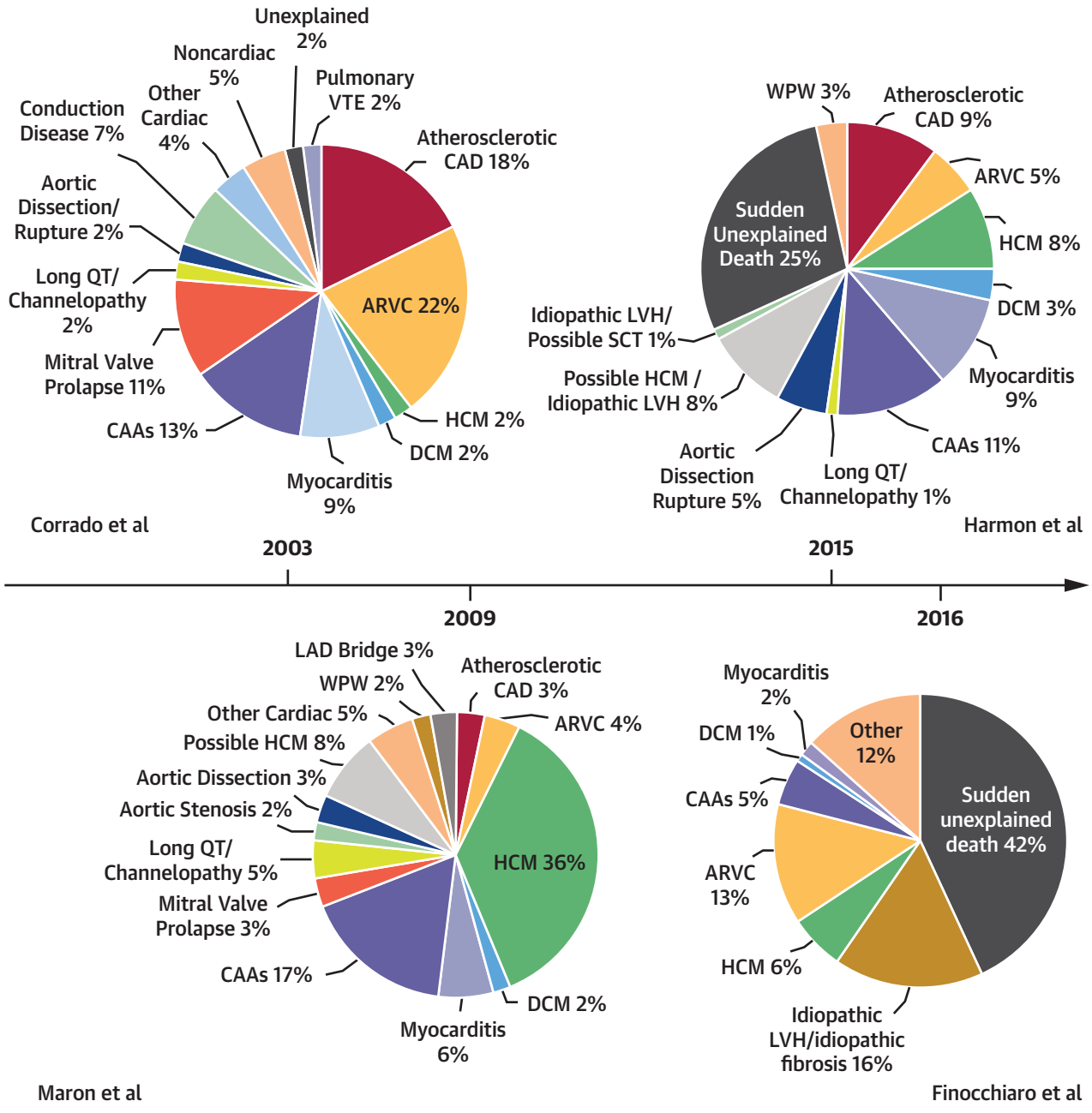
FIGURE 1 Causes of Sudden Cardiac Death in Athletes

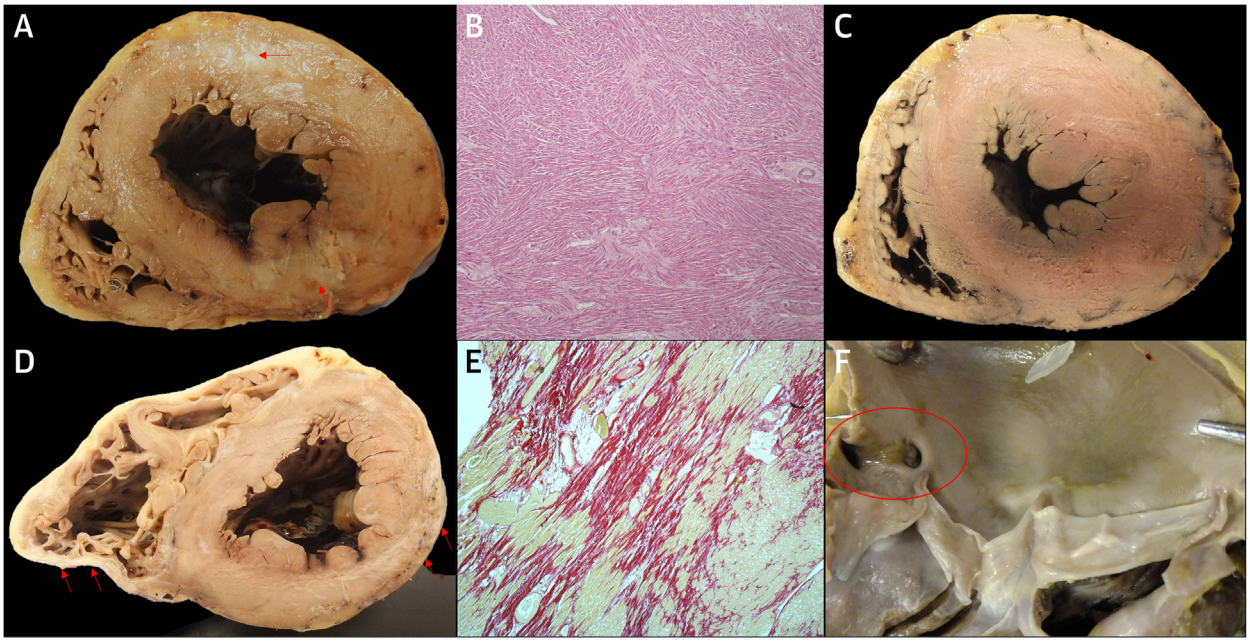


The attribution of specific conditions to sudden cardiac death in athletes has changed in the last 2 decades and is extremely variable according to methodology used, with some centers using a standardized protocol. ARVC = arrhythmogenic right ventricular cardiomyopathy; CAA = coronary anomaly; CAD = coronary artery disease; DCM = dilated cardiomyopathy; HCM = hypertrophic cardiomyopathy; LAD = left anterior descending artery; LVH = left ventricle hypertrophy; SCT = sickle cell trait; VTE = venous thromboembolism; WPW = Wolff-Parkinson-White.

detailed characterization of the origin and the course of the abnormal coronary artery. Although considerable debate remains regarding the optimal therapeutic management of these patients, surgical repair and coronary deroofing is generally preferred particularly if the left coronary artery arises from the right coronary sinus.[45]

Myocardial bridging (MB) is a congenital coronary anomaly characterized by the presence of a band of cardiac muscle overlying a segment of a coronary

FIGURE 2 Cardiac Autopsy Findings in Athletes Who Died Suddenly



(A) Macroscopic appearance of hypertrophic cardiomyopathy. Note the asymmetric hypertrophy and patchy fibrosis (red arrows). (B) Microscopic appearance of hypertrophic cardiomyopathy showing florid myocyte disarray. (C) Macroscopic appearance of idiopathic left ventricular hypertrophy. This may mimic hypertrophic cardiomyopathy but there is no myocyte disarray on histology. (D) Macroscopic appearance of arrhythmogenic cardiomyopathy. This biventricular arrhythmogenic cardiomyopathy case shows fat and fibrosis (red arrows) infiltrating into both left and right ventricles in a subepicardial distribution. (E) Microscopic appearance of arrhythmogenic cardiomyopathy. Fat and fibrosis infiltrating into the myocardium. (F) Anomalous right coronary artery. Note both ostia arising in the left coronary sinus (red circle).

artery. The left anterior descending coronary artery is the most frequently affected vessel.[46] This anomaly is relatively common in otherwise healthy individuals. In a large series of SCDs from the United Kingdom, MB was described in 0.5% of cases.[47] A left anterior descending coronary artery MB was reported in 3% of young athletes who died suddenly in the United States.[2]

ELECTRICAL CARDIAC ABNORMALITIES. Sudden deaths caused by electrical cardiac abnormalities are associated with a structurally normal heart at autopsy. Diagnosis can be made before death or following a cardiac arrest, typically after comprehensive investigations.

Long QT syndrome. Long QT syndrome (LQTS) is characterized by prolongation of the QT interval and propensity to ventricular arrhythmias. Polymorphic ventricular tachycardia, and specifically torsades de pointes and ventricular fibrillation, is one of the most feared complications. Deaths can be triggered by several stimuli affecting athletes, such as adrenergic surges associated with physical and emotional stress, electrolyte disturbances, and performance enhancing agents. Inheritance follows an autosomal dominant or rarely an autosomal recessive pattern.[48] LQTS has an estimated prevalence of up to 1:2,000.[49] Causative genes involved in the pathogenesis of LQTS encode potassium (KCNQ1 and KCNH2), sodium (SCN5A), and calcium (CALM1-3, CACNA1C) channels.[50] Although a report from an American registry attributed 3% of all SCDs to be caused by LQTS in young athletes, the true contribution of this condition to sudden fatal events is poorly understood.[2] A normal heart at autopsy of young athletes who died suddenly, which is suggestive of sudden arrhythmic death syndrome, is an increasingly common finding,[2,10] and LQTS is a possible cause of death in this setting. The diagnosis of LQTS relies on the assessment of the QT interval corrected for heart rate (QTc) and on a series of other ECG parameters measured at baseline and during exercise as well as features obtained by family and personal history.[50]

Brugada syndrome. Brugada syndrome (BrS) is an inherited cardiac ion channel disorder, most commonly affecting the sodium channels, and is

characterized by a propensity for ventricular arrhythmias and SCD. The diagnosis of BrS relies on the presence of the "type 1 Brugada pattern," which consists of partial right bundle branch block, ≥0.2 mV J-point elevation, and coved ST-segment elevation followed by T-wave inversion in the right ventricular leads of the 12-lead ECG.[51] This pattern can be spontaneous, or induced by sodium ion channel blocking drugs or transient conditions, such as high (>38 °C) body temperature. Inheritance follows an autosomal dominant pattern with a prevalence of 1:2,000. SCD in BrS often occurs at rest or during sleep. However, hyperthermia that may result from intense exercise and enhanced vagal tone, which is a common feature of physiological adaptation to exercise, may promote the development of fatal arrhythmias in athletes.[52]

Catecholaminergic polymorphic ventricular tachycardia. Catecholaminergic polymorphic ventricular tachycardia is an inherited arrhythmia syndrome characterized by stress-induced bidirectional ventricular tachycardia that may degenerate into ventricular fibrillation in the absence of structural cardiac abnormalities.[53] Although the baseline 12-lead ECG is often normal, multifocal ventricular premature beats or VT with beat to beat 180° alternating QRS axis (bidirectional VT) can be observed at the exercise tolerance test (ETT). Catecholaminergic polymorphic ventricular tachycardia most commonly results from mutations in 2 genes, RYR2 (cardiac ryanodine receptor RYR2—with an autosomal dominant line of inheritance) and CASQ2 (calsequestrin gene—with an autosomal recessive line of inheritance).[54]

Wolff-Parkinson-White. The combination of ventricular pre-excitation on the ECG and cardiac symptoms (typically palpitations and syncope) and/or paroxysmal supraventricular arrhythmias defines Wolff-Parkinson-White syndrome.[55] The prevalence of pre-excitation (delta wave and short PR interval) is estimated between 0.1% and 0.3%.[55,56] The risk of malignant arrhythmias in asymptomatic individuals is generally low, and ablation of the accessory pathway may decrease the risk of SCD in this condition.[57] A refractory period of <250 ms and inducibility of supraventricular tachycardia at electrophysiological study are indications for ablation.

OTHER CARDIAC CONDITIONS. Valvular heart disease. Mitral valve prolapse (MVP) is common in the general population (up to 2%-3% of cases).[58] In some rare cases, MVP may be associated with malignant arrhythmias. Several features can suggest an arrhythmic phenotype, including family history of SCD, T-wave inversion in the inferior leads, QTc interval prolongation at the ECG, ventricular extrasystoles arising from the mitral valve annulus (conducting with wide right bundle branch block and superior axis), mitral valve annular disjunction, bileaflet prolapse, and scarring affecting posterior papillary muscle and the adjacent inferior or posterior LV wall. Athletes with MVP should be evaluated with an exercise test, 24-hour ambulatory monitoring, and CMR. The demonstration of ventricular arrhythmias and the aforementioned high-risk features are red flags for athletes who wish to compete or engage in high intensity exercise.[59]

The significance of bicuspid aortic valve (BAV) in the context of SCD in poorly understood.[1,60] A recent study identified a BAV in 1% of cases of SCD, and death was attributed to aortic valve or aortic disease in one-half of them, whereas in the other half, the BAV was deemed to be an incidental finding.[61]

Aortic dissection. Type A aortic dissection is a rare cause of SCD in athletes. Remodeling of the aortic root may occur in athletes as a consequence of hemodynamic overload associated with exercise training. An aortic root dimension >40 mm in elite male athletes (and >34 mm in female athletes) is uncommon and is unlikely to represent the physiological consequence of exercise training.[62,63] Static exercise and typically weight training may result in acute rises in blood pressure to over 300 mm Hg,[64] with potential implications for aortic dilatation. Marfan syndrome, which is characterized by tall stature and may affect basketball or volleyball players, is a collagen disorder that is characterized by cardiac, ocular, and skeletal abnormalities and typically aortic dilatation. Affected individuals may present aortic dilatation and should be discouraged from engaging in intense static exercise that can increase the risk of aortic rupture or dissection.[35]

Commotio cordis. Commotio cordis is defined as SCD triggered by a sudden blow to the precordium. It is more common in children and adolescents who have pliable chest walls.[25,43] A series of experimental models indicate that if delivered at the onset of the T-wave in the cardiac cycle, precordial blows can result in ventricular fibrillation and cardiac arrest. In a recent study from the United Kingdom on a large cohort of adolescents who died suddenly, commotio cordis was the attributed cause of death in 1% of the cases and it was more common in athletes (5% compared with 1% in nonathletes).[43]

Performance enhancing drugs. The use of performance enhancing drugs is increasingly reported in athletes.[65] Doping substances are of many kinds and are classified according to the World Anti-Doping Agency code.[65] Some of the most commonly used agents include anabolic-androgenic steroids (AAS), stimulants and oxygen-carrying modulators.

AAS may lead to LVH, premature atherosclerosis, hypertension, and myocardial infarction. Mortality among athletes who abuse AAS is estimated to be 6 times higher than in nonabusing athletes, and around 30% of these deaths can be attributed to cardiovascular causes.[66] Stimulants such as ephedrine may cause hypertension, tremor, and palpitations.[65] Recombinant human erythropoietin stimulates hemoglobin production resulting in an increase in hematocrit and better athletic performance. A recent prospective cross-sectional study of 2,488 healthy older adults found a strong association between erythropoietin and incident heart failure, coronary heart disease, stroke, mortality, and decline in estimated glomerular filtration rate.[67]

MOLECULAR AUTOPSY. Postmortem genetic testing, referred to as "molecular autopsy" constitutes an important part of the diagnostic work-up of unexplained SCD and familial evaluation. The diagnostic yield of pathogenic and likely pathogenic variants is reported to be between 12% and 27% across studies using sequencing or cardiac panels for electrical disorders and cardiomyopathy.[68,69] When combined with family screening, the yield increased to 39%.[68]

Postmortem genetic testing has also been used in structural disease, with pathogenic and likely pathogenic variants found in 32% of those with cardiomyopathy identified at autopsy and 2% to 26% of those with structural disease of uncertain etiology, such as idiopathic hypertrophy and/or fibrosis.[69]

DIAGNOSTIC CHALLENGES IN ATHLETES

PHYSIOLOGICAL ADAPTATION TO EXERCISE: THE "ATHLETE'S" HEART. Participation in systematic vigorous exercise is associated with a modest increase in LV wall thickness, and biventricular cavity size with preserved systolic and diastolic function, which are established manifestations of the "athlete's heart."[70,71] A proportion of male athletes reveal cardiac dimensions that overlap with the cardiomyopathies (**Figure 3**). The differential diagnosis between athlete's heart and cardiomyopathy is generally confined to men with a large body surface area who engage in endurance sports such as rowing, canoeing, long distance cycling, and long-distance running.[72,73] Most of these athletes reveal ventricular dimensions that resemble dilated cardiomyopathy and arrhythmogenic right ventricular cardiomyopathy, whereas few show a wall thickness which overlaps with hypertrophic cardiomyopathy. In contrast, a significant proportion of athletes of African or Afro-Caribbean origin participating in explosive sports such as football, soccer, and basketball reveal an LV wall thickness, which overlaps with morphologically mild hypertrophic cardiomyopathy (HCM).[74,75] Increased trabeculations and repolarization changes may overlap with LV noncompaction cardiomyopathy and ion channel diseases respectively.[76,77]

OVERLAP WITH HCM. Large cohort studies have revealed that 1.5% to 2.0% of White male athletes and up to 13.0% of athletes of African or Afro Caribbean origin exhibit a wall thickness of 12 to 15 mm, which constitutes the gray zone between physiological LV hypertrophy and morphologically mild HCM.[78,79] An LV wall thickness ≥15 mm is uncommon irrespective of ethnicity and should be considered to represent pathological LV hypertrophy. The differentiation from HCM is based on the pattern of hypertrophy, concomitant LV cavity size, indexes of diastolic function, presence of LV outflow tract obstruction, myocardial scar within the hypertrophied segment, and repolarization abnormalities, typically lateral T-wave inversion. Most markers of diastolic function exhibit poor sensitivity for the diagnosis of HCM in an athlete. The most sensitive markers include a septal E <10 cm/s and a lateral E <12 cm/s.[80] CMR may be of value in the differential diagnosis, because of its high accuracy in wall thickness assessment and through the ability to reveal myocardial fibrosis, which is suggestive of HCM, rather than physiological adaptation. Although the diagnosis of HCM is based on cardiac imaging, the 12-lead ECG is an effective differentiator. In summary, among athletic subjects with an LV wall thickness of 12 to 15 mm, an asymmetric pattern of LVH, an LV cavity size <51 mm, a septal E <10 cm/s, and/or a lateral E <12 cm/s, an abnormal ECG consisting of T-wave inversion in the lateral leads, ST-segment depression, or pathological Q waves or myocardial scar within the hypertrophied segment at CMR support a diagnosis of HCM.

OVERLAP WITH ACM. A diagnosis of ACM may have significant implications in athletes. Intense exercise has been shown to worsen phenotypic expression and increase the propensity for fatal arrhythmias. The right ventricle (RV) is often dilated in athletes, resulting in a significant challenge in differential diagnosis. Dilatation of the RV is usually balanced in athletes, with RV end-diastolic volumes/LV end-diastolic volumes rarely exceeding a ratio of 1.2.[81] A suspicion of ACM should not be merely confined to the assessment of RV size and systolic function, but also on the presence or regional wall motion abnormalities, such as akinesia, dyskinesia, or aneurysm, and subepicardial or mid-wall fibrosis at CMR (**Figure 4**). The ECG may reveal "red flags" that point toward a diagnosis of ACM.[82] Anterior T-wave

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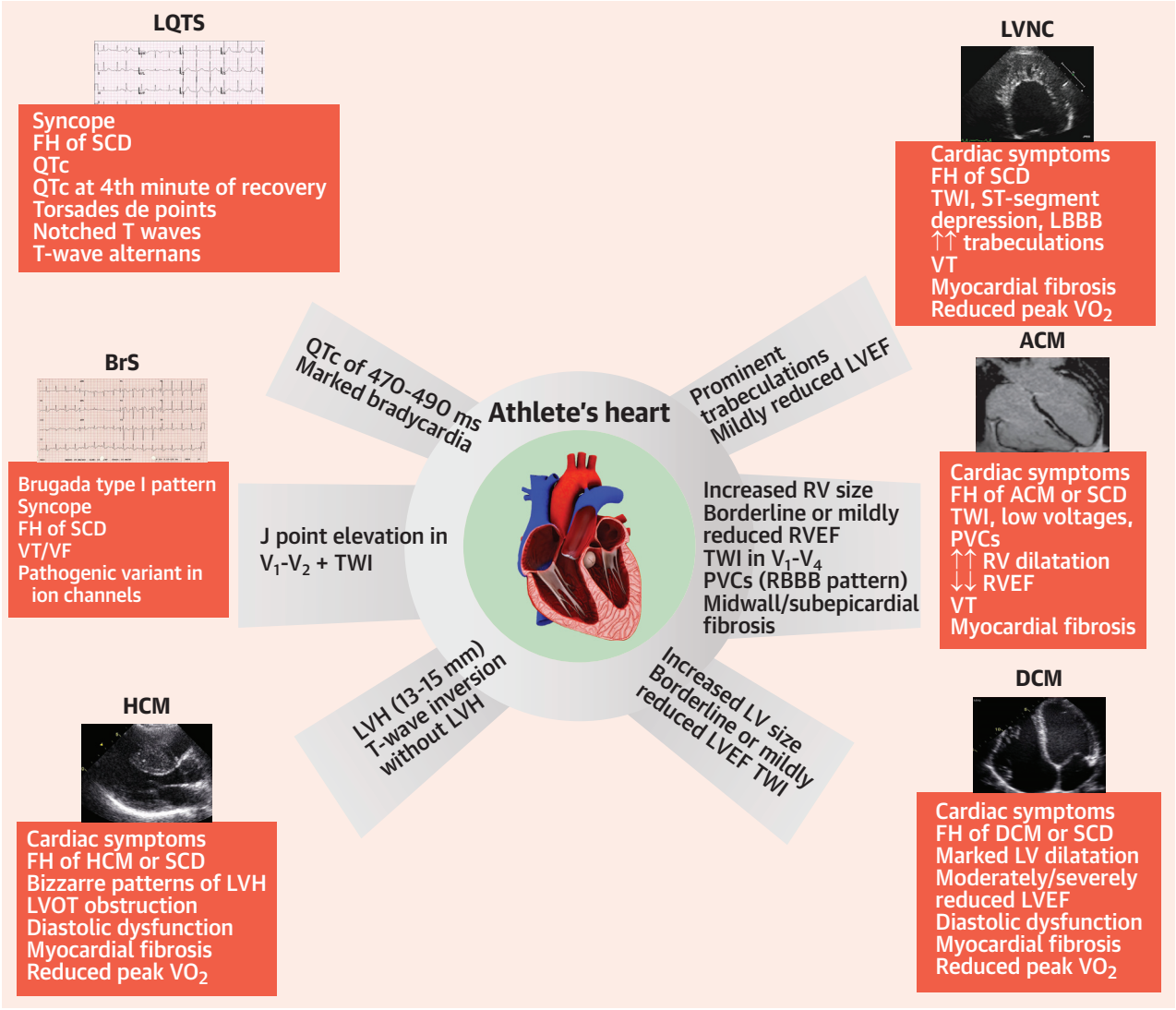
FIGURE 3 Overlap Between "Athlete's Heart" and Predominant Causes of SCD



Physiological electrical, structural, and functional changes in athletes may overlap with cardiac disease with a gray area that often needs clarification through second line cardiac investigations. ACM = arrhythmogenic cardiomyopathy; BrS = Brugada syndrome; DCM = dilated cardiomyopathy; FH = family history; HCM = hypertrophic cardiomyopathy; LQTS = long QT syndrome; RV = right ventricle; LVNC = left ventricular noncompaction; LVOT = left ventricular outflow tract; MV = mitral valve; RVEF = right ventricular ejection fraction; SCD = sudden cardiac death; TWI = T-wave inversion; VF = ventricular fibrillation; VT = ventricular tachycardia.

inversion is present in up to 50% of patients with ACM, which makes it the most common electrocardiographic features in ACM. Anterior T-wave inversion may be a marker of physiological adaptation to exercise in some asymptomatic athletes, particularly men of Afro-Caribbean origin in whom T-wave inversion in leads V_1 to V_4 is usually accompanied by preceding J-point elevation.[78,82]

OVERLAP WITH DILATED CARDIOMYOPATHY. Highly trained athletes may exhibit significant LV remodeling, which resembles a DCM phenotype. In Olympic athletes, LV size is higher than normal and almost one-half of the cases show an LV end-diastolic diameter >54 mm.[72] Athletes engaging in high dynamic intensity modalities, such as rowing, cross-country running, and cycling, often exhibit an LV end-diastolic diameter >60 mm.[83] Although highly trained athletes may exhibit an LV ejection fraction that is at the lower limits of normal or mildly reduced (45%-54%), moderate or severe LV systolic

impairment cannot be considered as a normal or benign feature. Our experience suggests that an increase in LVEF ≥11% from baseline to peak exercise at exercise echocardiography usually indicates physiological adaptation to exercise rather than a pathological process.[84]

OVERLAP WITH LV NONCOMPACTION. Left ventricular noncompaction (LVNC) is a cardiomyopathy characterized by prominent LV trabeculae and deep intertrabecular recesses.[85] Highly trained athletes may reveal some degree of LV trabeculation. In a large study of >1,000 asymptomatic athletes, 8.1% fulfilled conventional echocardiographic criteria for LVNC,[76] and this observation was more common in athletes of African/Afro-Caribbean origin. Unlike patients with LVNC, athletes usually have normal ECGs, preserved ventricular function, and absence of myocardial scar or exercise induced arrhythmias.[86]

OVERLAP WITH ION CHANNEL DISEASES. Some of the repolarization changes observed in highly trained athletes overlap with ECG changes associated with ion channel diseases, particularly the LQTS and Brugada syndrome.

The prevalence of a prolonged QTc interval in elite young athletes has been reported as high as 1 in 250 (0.4%).[87] The higher prevalence of a long QT interval in athletes may be caused by an increased cardiac mass or a consequence of autonomic adaptation. The measurement of the QTc interval in young athletes may be challenging as a result of training-related ECG changes. The Bazett's formula, which is commonly used to correct the QT intervals for heart rate variation, overestimates the QTc interval at profoundly low heart rates (<40 beats/min). Sinus arrhythmia may also result in overestimation of the QTc interval if the QT and R-R interval are not averaged for the calculation of the QTc. U waves are a common feature in young athletes, and erroneous inclusion of these in the QTc interval calculation can also overestimate the QTc interval.[88]

The international recommendations for ECG interpretation in athletes suggest a cutoff value of ≥470 ms in male athletes and ≥480 ms in female athletes for the potential diagnosis of LQTS.[89] Whereas these revised values provide a better false-positive and false-negative balance, they have a positive predictive value of <5% in isolation. It is our experience that a QTc interval ≥500 ms is more likely to be associated with other phenotypic features of the condition and is highly indicative of the diagnosis even in asymptomatic athletes. For asymptomatic athletes who have QTc intervals between 470 and 500 ms, further evaluation is recommended. The probability of having LQTS may be calculated using the modified Schwartz score, which is derived from the 12-lead ECG, personal and family history, Holter monitoring, and exercise stress testing. In addition to a long QT interval, the presence of notched T waves in 3 or more leads should raise suspicion of LQTS. Paradoxical prolongation of the QT interval during exercise does not feature in the Schwartz score but is a characteristic feature of long QT syndrome 1 and 2. Paradoxical prolongation can be usually demonstrated from a resting heart rate up to a heart rate of 120 beats/min, but it is extremely challenging to obtain an accurate measurement of the QT interval at higher heart rates.[90] Ideally, measurement of the QT interval should be performed in lead II or V_5 and should avoid leads V_2 to V_4, which have a high prevalence of U waves. The presence of polymorphic ventricular tachycardia during exercise or prolonged ECG monitoring is an uncommon finding, but it is a strong indicator of the disease. Genetic testing has a yield of approximately 70% and should be considered in equivocal cases with a high index of suspicion.[91]

It is important to differentiate the Brugada type 1 ECG pattern from other conditions associated with ST-segment elevation and r' in V_1 to V_2 including the right precordial early repolarization pattern commonly seen in trained athletes of African/Afro-Caribbean athletes. Early repolarization in the anterior leads in athletes is usually associated with "up-sloping" convex ST-segments after the J-point, whereas the ST-segment in BrS is "down-sloping" after the J-point. Measuring the ST segment at the J-point (STJ) and 80 ms after the J-point (ST80) may be helpful in this setting. In trained athletes, anterior early repolarization produces an STJ/ST80 ratio <1 because of the initial up-sloping of the ST-segment in this group.[89]

The type 2 pattern is characterized by a saddleback appearance in V_1 to V_2 with J-point elevation ≥2 mm and may also be a normal variant in athletes. The detection of this ECG pattern should lead to a repeat ECG with leads V_1 and V_2 placed in the second and third intercostal leads, which is more sensitive for the detection of the type 1 BrS pattern.

GENETIC TESTING IN ATHLETES. The diagnosis of inherited cardiac diseases is generally based on clinical investigations, which have a higher diagnostic yield than genetic testing itself. Genetic testing should be offered to individuals with overt clinical evidence of a genetic cardiac disorder to facilitate cascade screening for first-degree relatives. Next-generation sequencing has offered major advances in the molecular genetics of inherited cardiac diseases

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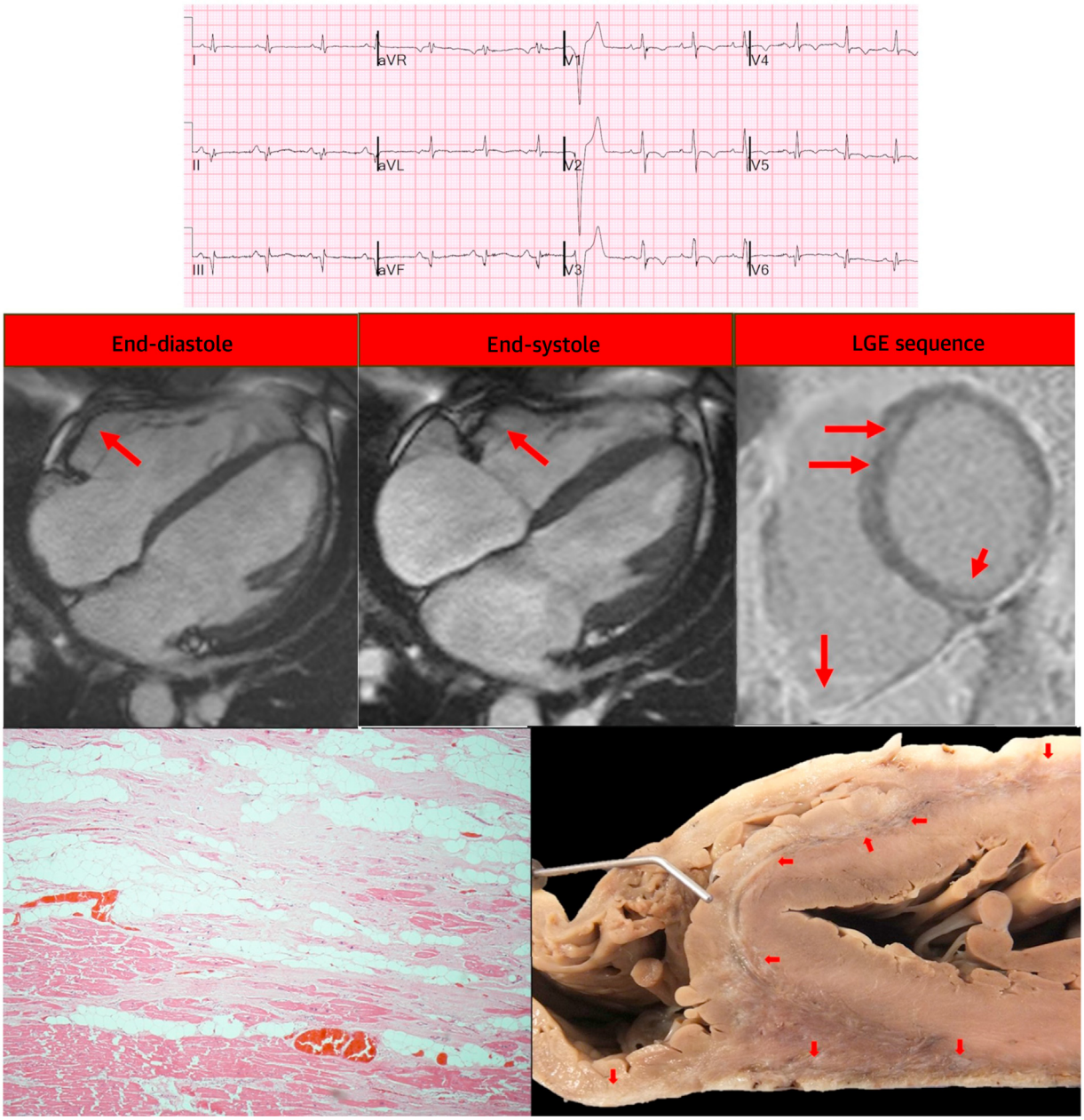
FIGURE 4 Case of an Athlete Who Died Suddenly



This figure shows the ECG, cardiovascular magnetic resonance, and postmortem pathology specimens of a 33-year-old athlete (footballer) with arrhythmogenic cardiomyopathy. The electrocardiogram shows widespread T-wave inversion and low voltages. Note the subtricuspidal aneurysm (in end-diastole and end-systole, red arrows) and the widespread mid-wall fibrosis, mainly within the interventricular septum and at the right ventricular free and inferior wall (red arrows) at cardiovascular magnetic resonance. The athlete died suddenly during a training session. At the postmortem examination, the macroscopic picture shows a streak of fibrosis (red arrows) through the septum that extends onto the epicardial surface of the free wall. There is fibrosis present in the posterior right ventricular free wall. Microscopically fibrosis and fat are seen infiltrating from the epicardial surface. LGE = late gadolinium enhanced.

and the development of relatively inexpensive genetic panels to investigate a broad spectrum of recognized mutations in genes implicated in cardiomyopathy and ion channel disorders, and has lowered the threshold of offering genetic testing to athletes in the gray zone between physiological cardiac enlargement and cardiomyopathy. In such instances, a genetic diagnosis may provide additional

prognostic information about the risk or progressive deterioration of cardiac function or arrhythmogenic SCD such as with pathogenic variants in genes encoding desmosomal proteins or lamin A/C.[92]

PREVENTION

SCD in athletes may be prevented through cardiac screening in apparently healthy individuals and through the implementation of policies that increase the likelihood of successful resuscitation of cardiac arrests.[93]

PREPARTICIPATION CARDIAC SCREENING. The American Heart Association (AHA)/American College of Cardiology (ACC) and the European Society of Cardiology recommend preparticipation cardiac screening to identify high-risk athletes.[35,94] However, there is no global consensus on whether or how cardiac screening should be performed, with high variability among countries, sports governing bodies, and level of competition. Both American and European guidelines suggest that medical history and physical examination should be part of the cardiac screening assessment.[35,94] Although the 12-lead ECG is the key investigation proposed by the European Society of Cardiology (**Figure 5**), this test is not recommended by the AHA/ACC, which focuses on personal and family history and physical examination. The AHA 14-point history and physical examination approach has low sensitivity because most athletes harboring potentially serious cardiac diseases are asymptomatic and do not exhibit abnormal physical signs.[95] The ECG is effective for identifying electrical disorders such as the Wolff-Parkinson-White ECG pattern and a prolonged QT interval. It is also well established that most individuals with cardiomyopathy reveal an abnormal ECG.[23,96,97] Corrado et al[98] showed that the implementation of a mandatory preparticipation cardiac screening program with the use of the ECG led to a significant decrease in the incidence of SCD in athletes: in the Veneto region of Italy, the annual incidence of SCD in athletes decreased by 89% after cardiac screening become compulsory in 1982. In contrast, Maron et al[99] analyzed 13 cases of SCD that occurred in high school student athletes and reported that only 4 (31%) individuals had cardiovascular conditions that could have been reliably detected through cardiac screening with history, examination, and ECG. Furthermore, Steinvil et al[100] showed that the effect of implementing mandatory preparticipation cardiac screening with ECG and exercise testing in Israel in 1997 did not reduce SCD events in competitive athletes. This study relied on the systematic search of 2 major newspapers in Israel, which constitutes a significant limitation because data were not collected through a prospective database, resulting in a possible underestimation of the events.[93] Some sports organizations complement ECG screening with echocardiography; however, the additional independent contribution of imaging in individuals with a normal ECG and the cost-effectiveness of this approach are unclear.[12]

Although preparticipation screening may be ethically justified to prevent SCD in athletes, there are a number of complex ethical issues relating to disqualification decisions.[101] Athletes, especially those competing at very high levels, may perceive disqualification as discrimination on the basis of a medical condition. Moreover, disqualification can cause significant psychological stress in athletes, which may affect their well-being in the long term. The use of mass cardiac screening in athletes remains controversial. The most cited issues are those relating to the economic sustainability of a nationwide screening program, the uncertain benefits in terms of SCD reduction, the potential for false positives and consequential unnecessary disqualifications, and the fact that screening does not prevent all cardiac deaths among young athletes.[101]

CARDIOPULMONARY RESUSCITATION. Immediate availability of quality cardiopulmonary resuscitation (CPR) performed by bystanders and automatic external defibrillators (AEDs) are crucial elements of SCD prevention. The greatest determinant of survival after a cardiac arrest is the time from collapse to defibrillation, with survival rates declining from 7% to 10% per minute for every minute lost.[102]

AEDs should be promptly available so that a first shock can be applied within the first minutes after the collapse. The role of education in the general population as well as the medical staff is key. The deployment of AEDs in public areas and training and sports grounds increases the chance of better outcomes. A recent investigation by Karam et al[103] described a substantial stability of incidence of sport-related SCA between 2005 and 2018 (the estimated incidence was 7.00 per million inhabitants/years in the first 2-year period of the study, compared with 6.24 per million inhabitants/years in the last 2 years). Despite this data, a significant improvement in survival to hospital discharge rates (23.8% in the first period compared to 66.7% in the last) was observed. This is probably a result of the increased education of the general population in terms of bystander CPR and use of AEDs. These results emphasize how important is to increase awareness of SCA in our society and how the delivery of appropriate CPR teaching

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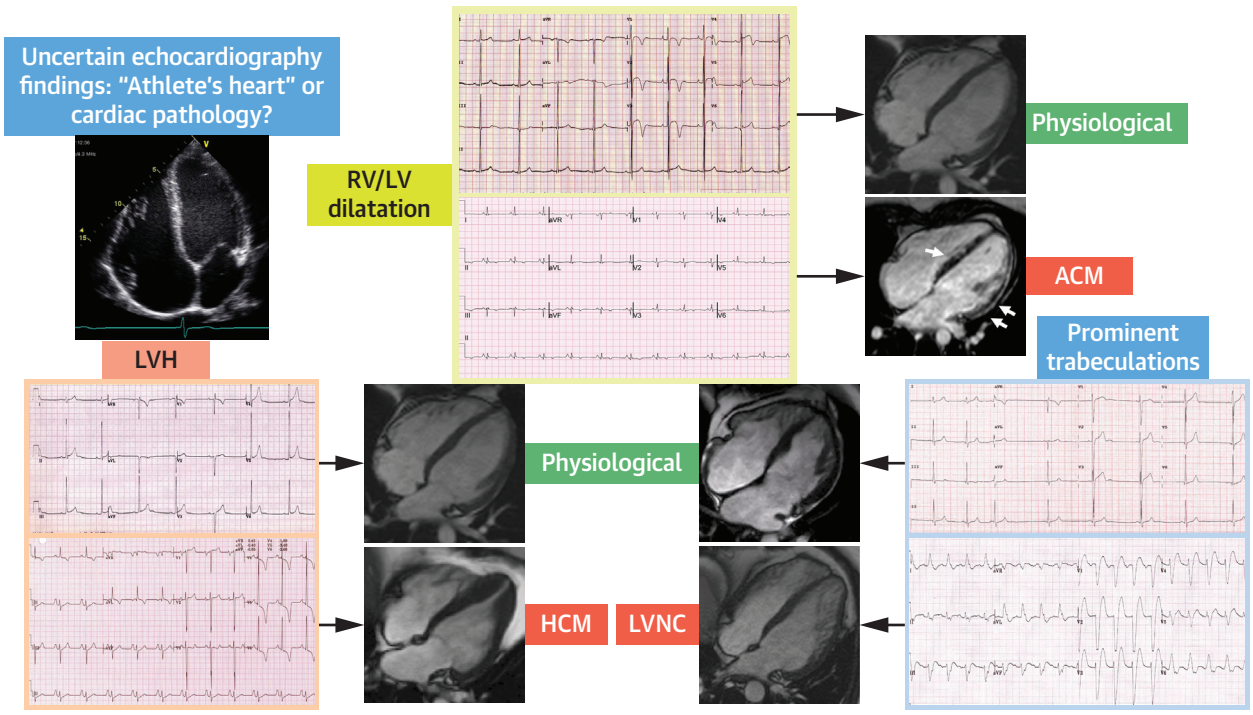
FIGURE 5 Clinical Dilemmas in the Differentiation Between "Athlete's Heart" and Cardiac Disease



The conundrum of equivocal findings at echocardiography (here we include right ventricular [RV] dilatation, prominent left ventricular [LV] trabeculations, and left ventricular hypertrophy [LVH]) may be clarified combining imaging and ECG findings. ACM = arrhythmogenic cardiomyopathy; HCM = hypertrophic cardiomyopathy; LVNC = left ventricular noncompaction.

programs in the general population may have a significant impact on outcomes.

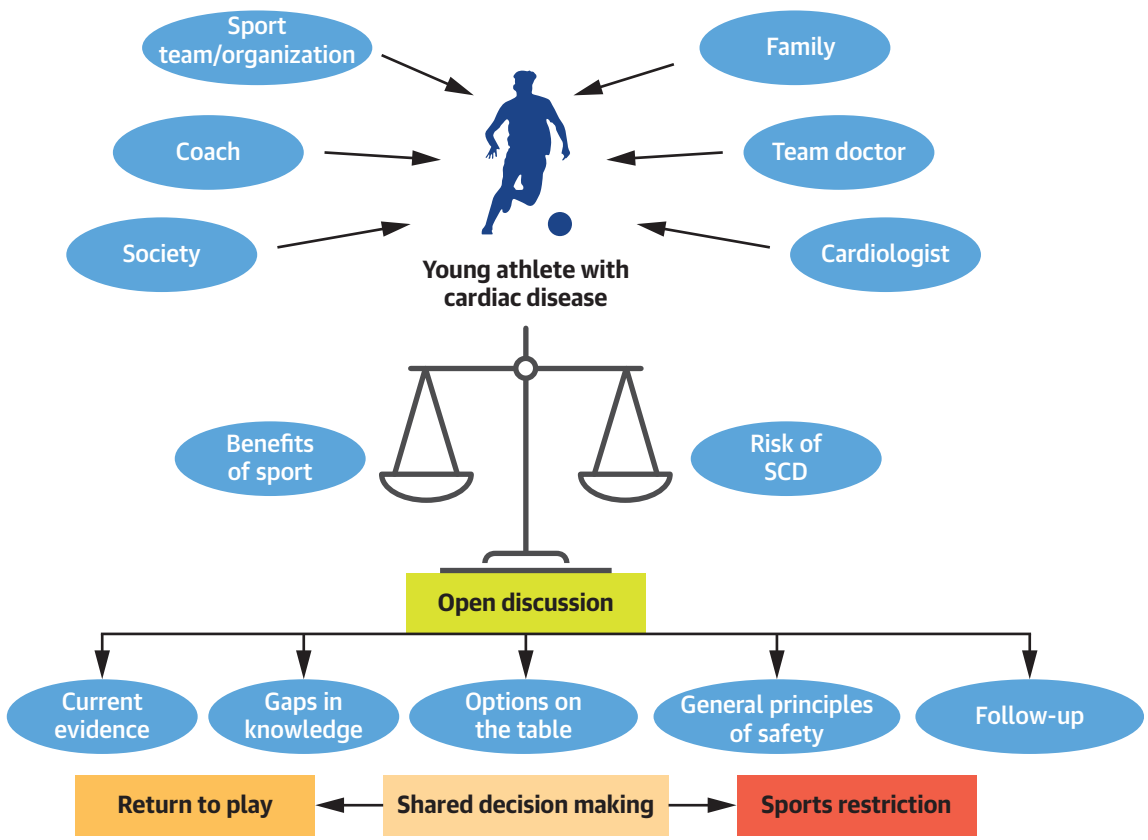
EXERCISE IN YOUNG ATHLETES WITH CARDIAC DISEASE

Because most athletes who die suddenly have underlying cardiac disease, uncertainties exist regarding risk and benefits of exercise in certain cardiovascular conditions. Robust evidence on the impact of different intensity and training volumes on risk of SCD in patients with overt cardiomyopathies or channelopathies is not available. International guidelines, which were traditionally restrictive for sports participation in athletes with cardiac disease, have embraced a more liberal attitude recently and promote a shared decision-making approach in absence of clinical markers of increased risk (**Figure 6**, **Table 2**).

According to previous recommendations, competitive exercise was categorically prohibited in the whole spectrum of patients with HCM. Over the past decade, there have been several observations indicating that physical activity and exercise of moderate intensity should be promoted in patients with HCM deemed at low risk. The heterogeneous morphology and pathophysiology of HCM means that some individuals are capable of participating in vigorous exercise including high-intensity competitive sports.[104,105]

An Italian study showed no excess of adverse events in athletes with HCM who continued to exercise, even at a high intensity level.[104] Data from the first multinational sports safety ICD registry report, which included 65 individuals with HCM, showed no adverse events during a 31-month follow-up period.[104] Although more individuals received appropriate and inappropriate shocks during physical activity than during resting conditions, there was no difference in shock rate between usual physical activity and competitive sport. This study suggests that even individuals deemed to be at high risk may be able to participate in vigorous exercise with an ICD in situ. The study was underpowered to assess the true impact of vigorous exercise, and although dynamic sports involving explosive sprints such as basketball

FIGURE 6 Exercise in Individuals With Cardiac Disease and Shared Decision Making



An athlete who is diagnosed with cardiac disease may be under pressure from various stakeholders, who often have a wide spectrum of opinions and ideas regarding the best course of action in terms of sports participation. Open discussion on the current evidence and gaps in knowledge in this field is crucial to embrace a shared decision making where the athlete is empowered to make the right decision. SCD = sudden cardiac death.

and soccer were included, few athletes participated in sports with high risk of collision such as American football or ice hockey, where the risk of damaging the generator box during collision may be higher. The prospective registry, "Lifestyle and Exercise in HCM" (LIVE-HCM), which included >600 exercising individuals with HCM showed no increased risk of SCD or life-threatening arrhythmias among individuals with HCM who exercised vigorously in comparison with those exercising moderately or those who were sedentary.[106]

The most contemporary sports cardiology guidelines and also the recent AHA/ACC guidelines for the management of patients with HCM are largely harmonized and recommend that among asymptomatic adults with morphologically mild HCM, particularly those who are habitually active and engaging in vigorous exercise at the time of diagnosis, competitive sport involving high-intensity effort may be considered after a detailed and transparent discussion that outlines the rare occurrence of adverse cardiac events but that also emphasizes the knowledge gaps in risk stratification and the unpredictability of SCA in athletes,[104,107] This precaution is especially relevant to specific high-risk populations such as adolescent and young adult men competing in sports requiring frequent bouts of explosive activity and young male athletes of African or Afro-Caribbean origin. This is because most exercise-related SCDs from HCM occur in adolescent and young adult male athletes involved in dynamic explosive sports such as basketball and soccer, with Black ethnicity conferring a higher risk. A select group of individuals with an ICD in situ may also engage in competitive sport after full discussion with an expert, although full contact/collision sport is not recommended.

Exercise appears to confer a higher risk of exercise-induced fatal arrhythmias and worsening of phenotype expression in ACM,[108] although available

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TABLE 2 Current AHA/ACC and ESC Recommendations for Sports Participation in Main Cardiomyopathies and Channelopathies

	2015 AHA/ACC Scientific Statements	2020 ESC Guidelines	2022 ESC Guidelines of VA and Prevention of SCD
Cardiomyopathies			
DCM	Symptomatic athletes with DCM—NO competitive sports, with the possible exception of low-intensity (class 1A sports). (III C)	Participation in low- to moderate-intensity recreational exercise should be considered in all individuals with DCM, regardless of the EF, in the absence of limiting symptoms, and exercise-induced VAs. (IIa C) Participation in high- or very high-intensity exercise including competitive sports (with the exception of those where occurrence of syncope may be associated with harm or death) may be considered in asymptomatic individuals who fulfil all of the following: 1) mildly reduced LV systolic function (EF 45%-50%); 2) absence of frequent and/or complex VAs on ambulatory Holter monitoring or exercise testing; 3) absence of LGE on CMR; 4) ability to increase EF by 10%-15% during exercise; and 5) no evidence of high-risk genotype (lamin A/C or filamin C). (IIb C) Participation in all competitive sports may be considered, with the exception of carriers of high-risk mutations (lamin A/C or filamin C). (IIb C) Participation in high- or very high-intensity exercise including competitive sports is not recommended for individuals with a DCM and any of the following: 1) symptoms or history of cardiac arrest or unexplained syncope; 2) LVEF <45%; 3) frequent and/or complex VAs on ambulatory Holter monitoring or exercise testing; 4) extensive LGE (>20%) on CMR; or 5) high-risk genotype (lamin A/C or filamin C). (III C)	Patients with DCM and a LMNA mutation—NO competitive sports. (III C)
HCM	Athletes with a probable or unequivocal clinical expression and diagnosis of HCM (ie, with the disease phenotype of LVH)—NO competitive sports, with the exception of those of low intensity (class IA sports). (III C) Participation in competitive athletics, in the absence of LVH by 2-dimensional echocardiography and CMR, is reasonable, particularly in the absence of a family history of HCM-related sudden death. (IIa C)	Participation in high-intensity exercise or competitive sports can be considered on the basis of an individualized expert assessment if no additional risk markers are present. (IIb C) High-intensity exercise is contraindicated in the presence of any HCM risk markers.[a] (III C) Patients who have risk markers may participate in low- or moderate-intensity recreational exercise depending on individualized expert assessment. (IIb C) Genotype positive, phenotype negative HCM patients may be eligible for high-intensity or competitive sport. (IIb C)	Participation in high-intensity exercise may be considered for asymptomatic adult HCM patients without risk markers. (IIb C)
ACM	Athletes with a definite diagnosis of ARVC should not participate in most competitive sports, with the possible exception of low-intensity class 1A sports. (III C) Athletes with a borderline diagnosis of ARVC should not participate in most competitive sports, with the possible exception of low-intensity class 1A sports. (III C) Athletes with a possible diagnosis of ARVC (genotype positive, phenotype negative individuals only show 1 major criterion, formally belonging to this category[a]) should not participate in most competitive sports, with the possible exception of low-intensity class 1A sports. (III C)	All individuals can be considered for low-intensity exercise. (IIa C) Patients who have risk markers may participate in low- or moderate-intensity recreational exercise depending on individualized expert assessment. (IIb C) High-intensity exercise and competitive sports are not recommended. (III B) Genotype positive, phenotype negative ARVC patients are not recommended to undertake high-intensity exercise or competitive sport Participation in high-intensity recreational exercise/sports or any competitive sports is not recommended. (III B) Annual assessment should be considered for phenotypic features and risk stratification purposes. (IIa C)	Avoidance of high-intensity exercise is recommended in patients with a definite diagnosis of ARVC. (I B) Avoidance of high-intensity exercise may be considered in carriers of ACM-related pathogenic mutations and no phenotype. (IIb C)

Continued on the next page

evidence is mainly from retrospective studies. Therefore, international guidelines are fairly conservative in this setting, and high-intensity exercise and competitive sports are not recommended in this condition.

In athletes with LQTS, high-intensity exercise is contraindicated where there has been prior arrhythmogenic syncope or cardiac arrest and in those with QTc >500 ms, but can be considered in individuals with mild or absent phenotype who are asymptomatic and optimally treated with beta-blocker following individualized expert assessment.[35,109]

GENOTYPE POSITIVE PHENOTYPE NEGATIVE ATHLETES. Genotype positive phenotype negative (G+P−) individuals harbor a pathogenic or likely pathogenic variant without exhibiting a phenotypic

TABLE 2 Continued

	2015 AHA/ACC Scientific Statements	2020 ESC Guidelines	2022 ESC Guidelines of VA and Prevention of SCD
Channelopathies			
LQTS	For an athlete with either symptomatic LQTS or electrocardiographically manifest LQTS (ie, corrected QT interval >470 ms in men or >480 ms in women), competitive sports participation (except competitive swimming in a previously symptomatic LQT1 host) may be considered after institution of treatment and appropriate precautionary measures assuming the athlete has been asymptomatic on treatment for at least 3 months. (IIb C) It is reasonable for an asymptomatic athlete with genotype positive, phenotype negative LQTS to participate in all competitive sports with appropriate precautionary measures.[b] (IIa C)	Shared decision making should be considered regarding sports participation. Type and setting of sports (individual vs team), type of mutation, and extent of precautionary measures should be considered in this context. (IIa C) Participation in high-intensity recreational and competitive sports, even when on beta-blockers, is not recommended in individuals with a QTc >500 ms or a genetically confirmed LQTS. (III B)	NA
BrS	Competitive sports participation may be considered for an athlete with either previously symptomatic or electrocardiographically evident BrS assuming appropriate precautionary measures and disease-specific treatments are in place and that the athlete has been asymptomatic on treatment for at least 3 months. (IIb C) It is reasonable for an asymptomatic athlete with genotype positive, phenotype negative BrS to participate in all competitive sports with appropriate precautionary measures.[c] (IIa C)	Participation in sports activities that are not associated with an increase in core temperature >39°C (eg, endurance events under extremely hot and/or humid conditions) may be considered. (IIb C) Following implantation of an ICD, resumption of leisure or competitive sports should be considered after shared decision making in individuals who have not experienced recurrent arrhythmias over 3 months after ICD implantation. (IIa C)	NA

[a]Cardiac symptoms, unexplained syncope, or history of cardiac arrest, moderately elevated European Society of Cardiology (ESC) hypertrophic cardiomyopathy (HCM) risk score of 5-year sudden cardiac death (SCD) risk (≥4%), resting left ventricular outflow tract gradient >30 mm Hg, abnormal blood pressure response to exercise (failure of blood pressure to increase appropriately with exercise), and history of exercise-induced arrhythmias. [b]Includes the following: 1) avoidance of QT-prolonging drug; 2) electrolyte/hydration replenishment and avoidance of dehydration; 3) avoidance or treatment of hyperthermia from febrile illnesses or training-related heat exhaustion or heat stroke; 4) acquisition of a personal automatic external defibrillator as part of the athlete's personal sports safety gear; 5) establishment of an emergency action plan with the appropriate school or team officials. [c]Includes the following: 1) avoidance of drugs that exacerbate the Brugada syndrome (BrS); 2) electrolyte/hydration replenishment and avoidance of dehydration; 3) avoidance/treatment of hyperthermia from febrile illnesses or training-related heat exhaustion or heat stroke; 4) acquisition of a personal automatic external defibrillator as part of the athlete's personal sports safety gear; 5) establishment of an emergency action plan with the appropriate school or team officials.

ACM = arrhythmogenic cardiomyopathy; AED = automatic external defibrillator; AHA/ACC = American Heart Association/American College of Cardiology; ARVC = arrhythmogenic right ventricular cardiomyopathy; CMR = cardiac magnetic resonance; CPVT = catecholaminergic polymorphic ventricular tachycardia; DCM = dilated cardiomyopathy; EF = ejection fraction; LGE = late gadolinium enhancement; LQTS = long QT syndrome; LVEF = left ventricular ejection fraction; LVH = left ventricle hypertrophy; NA = nonapplicable; SQTS = short QT syndrome; VA = ventricular arrhythmia.

manifestation of the disease.[110] The knowledge on the effects of intense exercise in G+P− individuals is limited, and the general attitude is to oversimplify clinical management based on phenotypes. The recommendation to engage in intensive exercise and competitive sports is usually contingent on annual clinical surveillance. However, when actionable variants in specific genes, such as LMNA or PKP2, are found, guidelines do not differentiate between G+P− individuals and patients with overt disease, emphasizing the arrhythmic risk and recommending avoidance of participation in high-intensity recreational exercise and competitive sports.[35,111,112]

FUTURE PERSPECTIVES

SCD in young athletes draws many questions regarding the epidemiology, causes, and prevention of these catastrophes (**Central Illustration**). Implementation of regional or national registries where autopsy is performed by a dedicated cardiac pathology laboratory with expertise in this setting is crucial to provide more robust evidence on incidence and underlying etiologies. Gaps in knowledge regarding the impact of intensive exercise in individuals with cardiovascular diseases should be addressed through randomized clinical trials that may inform the

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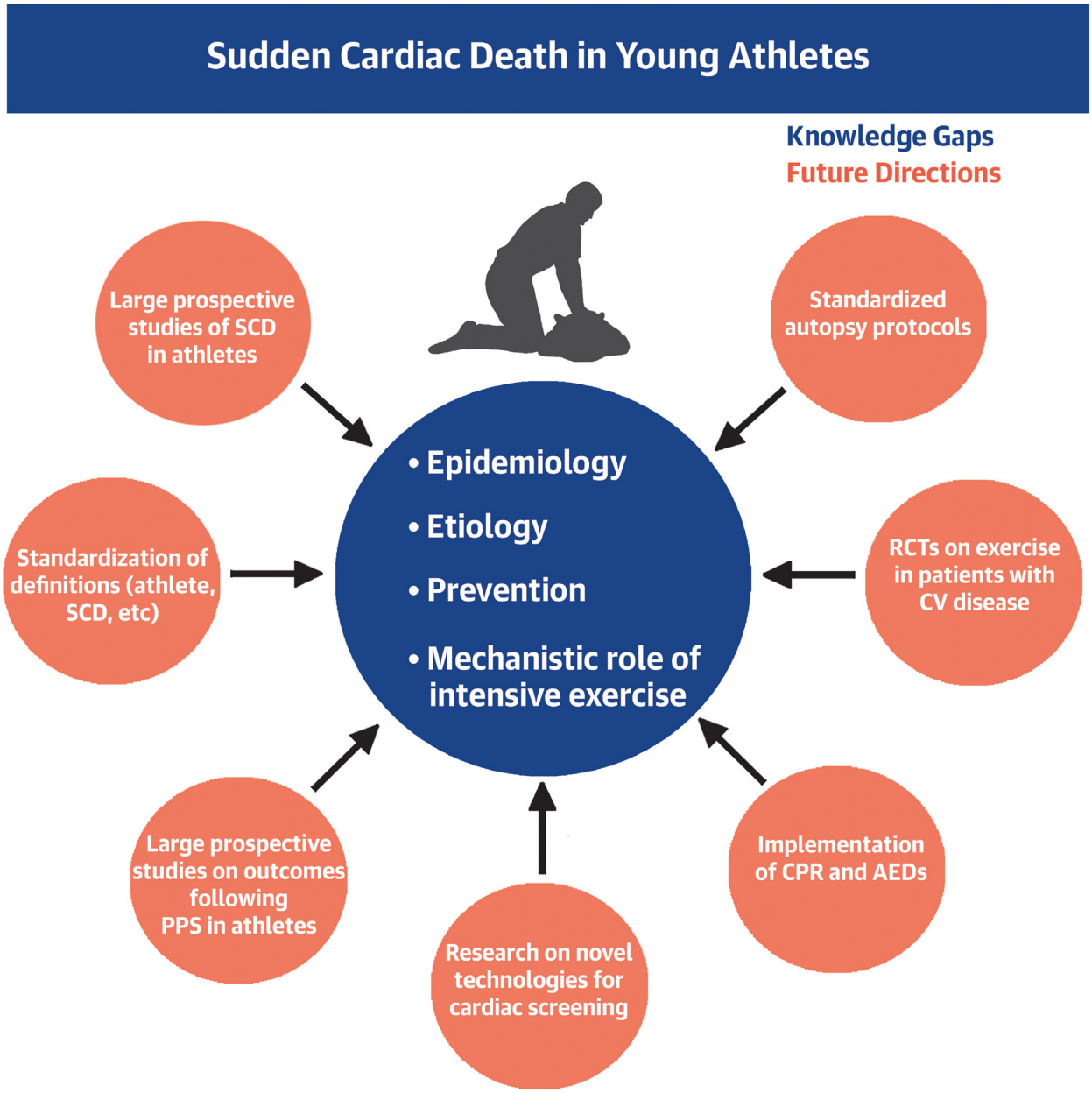
CENTRAL ILLUSTRATION **Gaps in Knowledge and Future Directions in Sudden Cardiac Death in Young Athletes**



Finocchiaro G, et al. J Am Coll Cardiol. 2024;83(2):350-370.

SCD in young athletes is a subject that remains controversial, with recent advances in the field, but still many gaps in knowledge (blue circle). Future directions (orange circles) include development of large prospective registries that may inform on epidemiology and the use of standardized autopsy protocols would increase the diagnostic accuracy. Preparticipation cardiac screening remains a matter of debate, with questions regarding its use in large cohorts. The mechanistic role of intense exercise on cardiac phenotype and its safety in individuals with cardiac disease is a territory that will need to be explored to provide tailored recommendations on exercise prescription. AED = automatic external defibrillator; CPR = cardiopulmonary resuscitation; CV = cardiovascular; PPS = preparticipation screening; RCT = randomized controlled trial; SCD = sudden cardiac death.

optimal safe dose of exercise in these patients. Preparticipation screening remains a means to detect early serious cardiac disease; however, theories on its beneficial role on hard endpoints (eg, prevention of SCD in large populations) remain speculative with major concerns on cost-effectiveness. Large-scale prospective longitudinal studies, perhaps using novel technologies (wearable devices or smartphone-based ECG) are necessary to corroborate the evidence in this context. Knowledge on determinants of ECG variability in athletes, including race, sex, age, and others, is limited to observational cross-sectional data, whereas longitudinal studies may provide further insight on their significance.

CONCLUSIONS

Sudden cardiac death in a young athlete is an uncommon but highly tragic event. The precise epidemiological burden of SCD is not well established, with a high variability in terms of incidence among different studies. Young men of African and African-American origin appear to be the population at the highest risk. A diverse spectrum of diseases is implicated in SCD in young athletes. Cardiac assessment of some athletes can be challenging when the electrical and structural changes consequent to physiological adaptation to exercise overlap with cardiac disease. A holistic interpretation of clinical findings in conjunction with important elements in personal and family history is crucial for resolving the diagnostic conundrum. Prevention can be achieved through preparticipation screening and implementation of policies aimed at prompt cardiopulmonary resuscitation. Exercise prescription in young individuals with potentially serious cardiac diseases should embrace shared decision making whilst acknowledging knowledge gaps in this setting.

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